

Large-Instance Computational Fragments

The displayed labels match the current manuscript labels only by reference. The blocks below are intended as add-on fragments, not automatic replacements for the existing tables.

Larger max-distance instances

We generated 30 additional max-distance instances using the same generator and rank-one Shor prefilter as in Tables ?? and ?. Full KRON, Full SEP, and Gurobi were omitted. The table columns therefore retain only the large-scale methods, Lazy KRON and Lazy SEP. In the diagnostics table, tight means that the relative gap between the relaxation bound and the objective value of the recovered first-column point is at most 10^{-5} . Relative rank-one residuals and recovered-point feasibility violations are also calculated and retained in the summary CSV, but neither enters the definition of tight.

Table 1: Results for larger max-distance instances

n	cases	Lazy KRON			Lazy SEP			Lazy KRON			Lazy SEP		
		time	tight	max gap	time	tight	max gap	rounds	cuts	max viol.	rounds	cuts	max skew
20	10	294 ms	8	$1.7 \cdot 10^{-5}$	1.01 s	10	$1.7 \cdot 10^{-7}$	4/9	3/8	$4.7 \cdot 10^{-8}$	1/2	0/50	$2.8 \cdot 10^{-8}$
25	10	3.85 s	9	$9.2 \cdot 10^{-5}$	3.20 s	10	$9.5 \cdot 10^{-8}$	3/7	2/6	$6.7 \cdot 10^{-8}$	1/1	0/0	$1.6 \cdot 10^{-9}$
30	10	14.54 s	9	$1.8 \cdot 10^{-5}$	9.17 s	10	$1.7 \cdot 10^{-8}$	4/7	3/6	$8.5 \cdot 10^{-8}$	1/3	0/100	$1.5 \cdot 10^{-9}$

Larger generated TTRS instances

We generated 30 centered affine-diagonal TTRS instances and used Shor only as a prefilter to discard instances already certified by rank-one recovery. The reported large-scale methods are Lazy KRON and Lazy SEP. Since no beta or Gurobi reference is used in this generated panel, tight means that the relative gap between a relaxation bound and the objective value of its recovered point is at most 10^{-5} . Relative rank-one residuals and recovered-point feasibility violations are also calculated and retained in the summary CSV, but neither enters the definition of tight.

Table 2: Results for additional TTRS instances

n	cases	Lazy KRON			Lazy SEP			Lazy KRON			Lazy SEP		
		time	tight	max gap	time	tight	max gap	rounds	cuts	max viol.	rounds	cuts	max skew
20	10	309 ms	4	$4.6 \cdot 10^{-5}$	1.01 s	10	$1.6 \cdot 10^{-7}$	4/18	4/17	$8.8 \cdot 10^{-8}$	1/5	0/100	$7.5 \cdot 10^{-8}$
25	10	1.47 s	6	$3.8 \cdot 10^{-1}$	3.69 s	10	$3.8 \cdot 10^{-6}$	7/22	6/21	$7.8 \cdot 10^{-8}$	1/10	0/900	$9.9 \cdot 10^{-8}$
30	10	20.14 s	4	$4.2 \cdot 10^{-1}$	12.15 s	10	$5.4 \cdot 10^{-7}$	6/37	5/36	$9.6 \cdot 10^{-8}$	2/6	52/500	$9.2 \cdot 10^{-8}$